

Problem on Multinomial Naive Bayes

Problem

- Suppose we have a small training corpus of 6 documents, labeled as Sports or Politics:
- Training Data:
- Sports: "goal goal win" (Doc1), "win match" (Doc2), "goal win" (Doc3)
- Politics: "election vote" (Doc4), "vote debate" (Doc5), "election debate" (Doc6)

Problem

- Vocabulary: All unique words = {goal, win, match, election, vote, debate}
- |Vocabulary| = 6
- Smoothing parameter: $\alpha = 1$ (Laplace smoothing)

Step 1: Calculate $P(c_j)$ - Prior probabilities

- Total documents = 6
- Sports documents = 3 $\rightarrow P(\text{Sports}) = 3/6 = 0.5$
- Politics documents = 3 $\rightarrow P(\text{Politics}) = 3/6 = 0.5$

Step 2: Create combined documents for each class

- Text_{or} = "goal goal win win match goal win"
- (All sports docs combined)
- Text_{o_i_ic} = "election vote vote debate election debate"
- (All politics docs combined)

Step 3: Count word frequencies in each class

- Sports:
- goal: 3 occurrences
- win: 3 occurrences
- match: 1 occurrence
- election: 0
- vote: 0
- debate: 0
- Total words in Sports text = $3+3+1 = 7$

Step 3: Count word frequencies in each class

- Politics:
- election: 2
- vote: 2
- debate: 2
- goal: 0
- win: 0
- match: 0
- Total words in Politics text = $2+2+2 = 6$

Step 4: Calculate $P(w_k | c_j)$ with Laplace smoothing

- Formula:

$$P(w_k | c_j) = \frac{n_k + \alpha}{N + \alpha \times |Vocabulary|}$$

- where:

- n_k = count of word w_k in class c_j
- N = total word count in class c_j
- $\alpha = 1$ (smoothing)
- $|Vocabulary| = 6$

Step 3: Count word frequencies in each class

- For Sports:
- $N = 7$, denominator = $7 + 1 \times 6 = 13$
- $P(\text{goal}|\text{Sports}) = (3 + 1)/13 = 4/13 \approx 0.3077$
- $P(\text{win}|\text{Sports}) = (3 + 1)/13 = 4/13 \approx 0.3077$
- $P(\text{match}|\text{Sports}) = (1 + 1)/13 = 2/13 \approx 0.1538$
- $P(\text{election}|\text{Sports}) = (0 + 1)/13 = 1/13 \approx 0.0769$
- $P(\text{vote}|\text{Sports}) = (0 + 1)/13 = 1/13 \approx 0.0769$
- $P(\text{debate}|\text{Sports}) = (0 + 1)/13 = 1/13 \approx 0.0769$

Step 3: Count word frequencies in each class

- For Politics:
- $N = 6$, denominator = $6 + 1 \times 6 = 12$
- $P(\text{election}|\text{Politics}) = (2 + 1)/12 = 3/12 = 0.25$
- $P(\text{vote}|\text{Politics}) = (2 + 1)/12 = 3/12 = 0.25$
- $P(\text{debate}|\text{Politics}) = (2 + 1)/12 = 3/12 = 0.25$
- $P(\text{goal}|\text{Politics}) = (0 + 1)/12 = 1/12 \approx 0.0833$
- $P(\text{win}|\text{Politics}) = (0 + 1)/12 = 1/12 \approx 0.0833$
- $P(\text{match}|\text{Politics}) = (0 + 1)/12 = 1/12 \approx 0.0833$

Learned Parameters

- Priors:
- $P(\text{Sports}) = 0.5$
- $P(\text{Politics}) = 0.5$

Learned Parameters

- Likelihoods ($P(\text{word}|\text{class})$):

Word	$P(\text{word} \text{Sports})$	$P(\text{word} \text{Politics})$
goal	0.3077	0.0833
win	0.3077	0.0833
match	0.1538	0.0833
election	0.0769	0.2500
vote	0.0769	0.2500
debate	0.0769	0.2500

Observations

- Without smoothing, words like "election" would have $P(\text{election}|\text{Sports}) = 0$, causing problems if a test document contained that word.
- With $\alpha=1$ smoothing, all words have non-zero probability.
- Sports documents have higher probability for sports-related words, and politics documents for politics-related words.
- The sum of $P(w|\text{Sports})$ for all words in vocabulary = 1 (and same for Politics), as required for a probability distribution.